

# OMRON E5CC

Digital Temperature Controller — 48 × 48 mm

|                       |                                                                          |
|-----------------------|--------------------------------------------------------------------------|
| Product family        | E5CC / E5CC-B / E5CC-U                                                   |
| Function              | General-purpose digital temperature controller                           |
| Control methods       | ON/OFF control or 2-PID control with auto-tuning                         |
| Input types           | Thermocouple, Pt100/JPt100, ES1B infrared sensor, analog current/voltage |
| Display               | 11-segment digital display; PV character height 15.2 mm, SV 7.1 mm       |
| Input sampling period | 50 ms                                                                    |
| Panel size            | 48 × 48 mm                                                               |

## Key Features

- Large white PV display for improved readability.
- Universal temperature input supporting a broad range of thermocouples and platinum resistance thermometers.
- Analog input options including 4–20 mA, 0–20 mA, 1–5 V, 0–5 V, 0–10 V and, for applicable E5CC-U models, 0–50 mV.
- Relay, SSR-driving voltage, and linear current output options depending on model.
- Heating/cooling control, multi-set-point operation, alarms, heater burnout detection, self-tuning and robust tuning.
- Up to eight set points can be stored and selected on applicable E5CC models.

## Ratings & Electrical Specifications

| Parameter                  | Specification                                                                                                     |
|----------------------------|-------------------------------------------------------------------------------------------------------------------|
| Power supply               | 100–240 VAC, 50/60 Hz (A models); 24 VAC, 50/60 Hz or 24 VDC (D models)                                           |
| Operating voltage          | 85% to 110% of rated supply voltage                                                                               |
| Power consumption          | Option 000: 5.2 VA max. at 100–240 VAC; 3.1 VA max. at 24 VAC; 1.6 W max. at 24 VDC. Other models: 6.5 VA / 4.1 W |
| Control output — relay     | E5CC/E5CC-B: SPST-NO, 250 VAC, 3 A resistive; E5CC-U: SPDT, 250 VAC, 3 A resistive                                |
| Control output — SSR drive | 12 VDC ±20%, PNP, 21 mA max., short-circuit protection                                                            |
| Linear current output      | 4–20 mA DC or 0–20 mA DC; 500 Ω max.; approx. 10,000 resolution                                                   |
| Auxiliary outputs          | E5CC: 3; E5CC-B: 2; E5CC-U: 1 or 2 depending on model                                                             |
| Event inputs               | E5CC: 2 or 4 depending on model; E5CC-B: 2 depending on model                                                     |
| Transfer output            | 4–20 mA, 500 Ω max., or 1–5 VDC, 1 kΩ min., depending on model                                                    |

## Input Specifications

| Input category          | Supported input                                          |
|-------------------------|----------------------------------------------------------|
| Thermocouple            | K, J, T, E, L, U, N, R, S, B, C/W, PL II                 |
| Platinum resistance     | Pt100, JPt100                                            |
| Infrared sensor         | ES1B: 10–70°C, 60–120°C, 115–165°C, or 140–260°C         |
| Current                 | 4–20 mA, 0–20 mA                                         |
| Voltage                 | 1–5 V, 0–5 V, 0–10 V; 0–50 mV on applicable E5CC-U units |
| Current input impedance | 150 $\Omega$ max.                                        |
| Voltage input impedance | 1 M $\Omega$ min.                                        |

## Analog Scaling

| Setting range        | Scaled usable range |
|----------------------|---------------------|
| General scaling      | –1999 to 9999       |
| One decimal place    | –199.9 to 999.9     |
| Two decimal places   | –19.99 to 99.99     |
| Three decimal places | –1.999 to 9.999     |

## Accuracy & Control Characteristics

| Parameter                         | Specification                                                                                                               |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| E5CC/E5CC-B thermocouple accuracy | ( $\pm 0.3\%$ of indication value or $\pm 1^\circ\text{C}$ , whichever is greater) $\pm 1$ digit max. at $23^\circ\text{C}$ |
| E5CC/E5CC-B Pt accuracy           | ( $\pm 0.2\%$ of indication value or $\pm 0.8^\circ\text{C}$ , whichever is greater) $\pm 1$ digit                          |
| E5CC/E5CC-B analog accuracy       | $\pm 0.2\%$ FS $\pm 1$ digit max.                                                                                           |
| E5CC-U thermocouple accuracy      | ( $\pm 1\%$ of indication value or $\pm 2^\circ\text{C}$ , whichever is greater) $\pm 1$ digit max.                         |
| Transfer output accuracy          | $\pm 0.3\%$ FS max.                                                                                                         |
| Remote SP input accuracy          | $\pm 0.2\%$ FS $\pm 1$ digit max.                                                                                           |
| Hysteresis — temperature          | 0.1–999.9°C/°F in 0.1°C/°F increments                                                                                       |
| Hysteresis — analog               | 0.01–99.99% FS in 0.01% FS increments                                                                                       |
| P / I / D ranges                  | P: 0.1–999.9°C/°F or 0.1–999.9% FS; I/D: 0–9999 s or 0.0–999.9 s                                                            |
| Control period                    | 0.1, 0.2, 0.5, or 1–99 s                                                                                                    |

## Environmental & Mechanical Specifications

| Parameter                | Specification                                                                                                 |
|--------------------------|---------------------------------------------------------------------------------------------------------------|
| Operating temperature    | –10 to 55°C; for 3-year warranty, –10 to 50°C with standard mounting                                          |
| Operating humidity       | 25–85% RH, no condensation or icing                                                                           |
| Storage temperature      | –25 to 65°C, no condensation or icing                                                                         |
| Maximum altitude         | 2,000 m                                                                                                       |
| Recommended fuse         | T2A, 250 VAC, time-lag, low-breaking capacity                                                                 |
| Installation environment | Overvoltage Category II; Pollution Degree 2                                                                   |
| Insulation resistance    | 20 MΩ min. at 500 VDC                                                                                         |
| Dielectric strength      | 3,000 VAC, 50/60 Hz for 1 min between terminals of different charge                                           |
| Vibration — malfunction  | 10–55 Hz, 20 m/s², 10 min each in X, Y and Z                                                                  |
| Vibration — resistance   | 10–55 Hz, 20 m/s², 2 h each in X, Y and Z                                                                     |
| Shock — malfunction      | 100 m/s², 3 times each in X, Y and Z                                                                          |
| Shock — resistance       | 300 m/s², 3 times each in X, Y and Z                                                                          |
| Weight                   | E5CC/E5CC-B controller approx. 120 g; E5CC-U approx. 100 g; mounting adapter approx. 10 g                     |
| Protection               | E5CC/E5CC-B: front IP66 / UL Type 1, rear IP20, terminals IP00. E5CC-U: front IP50, rear IP20, terminals IP00 |
| Memory                   | Non-volatile memory; 1,000,000 write operations                                                               |

## Functions

| Control / tuning  | ON/OFF, 2-PID, auto-tuning, 40% AT, 100% AT, self-tuning, robust tuning                               |
|-------------------|-------------------------------------------------------------------------------------------------------|
| Process functions | Heating/cooling, SP ramp, MV limiter, input digital filter, PV input shift, moving average            |
| Alarm functions   | 19 alarm types, heater burnout (HB), SSR failure (HS), loop burnout and alarm delay functions         |
| Operation         | Manual output, run/stop, protection, display brightness setting, logic operations, simple programming |
| Set points        | Up to 8 SPs (SP0–SP7) on applicable E5CC models; up to 4 when E5CC-B uses event inputs                |

## Standards & Connectivity

| Item                | Specification                                                                                                       |
|---------------------|---------------------------------------------------------------------------------------------------------------------|
| Approved standards  | cULus: UL 61010-1 / CSA C22.2 No. 61010-1; some models include KOSHA and KC; Lloyd's standards on applicable models |
| Conformed standards | EN 61010-1 (IEC 61010-1), RCM                                                                                       |
| EMC                 | EN 61326-1; EMI EN 55011 Group 1 Class A; EMS immunity per EN 61326-1                                               |
| Setup tool          | E5CC: CX-Thermo v4.5+; E5CC-B: v4.65+; E5CC-U: v4.61+                                                               |
| Setup port          | Top panel; E58-CIFQ2 USB-Serial Conversion Cable                                                                    |

Source: OMRON Industrial Automation — E5CC, E5CC-B, E5CC-U specifications and catalog. Official specification page last updated January 5, 2026; official data sheet listing currently shows H177-E1-19, updated August 3, 2026.

This PDF is a clean technical datasheet compiled from OMRON's published specifications. For ordering, wiring, model selection, installation and safety requirements, consult the applicable OMRON model-specific data sheet and user's manual.